

ECON 144 - Comprehensive Study Guide

1. Time Series Fundamentals - Definitions: stochastic process, time series, stationarity (weak/strong), autocorrelation. - Decomposition: Trend, Seasonality, Cycles, Irregular. - ACF/PACF interpretation. - Transformations: differencing, logs. - Tests: * Ljung-Box Test: H_0 : no serial correlation. * ADF Test: H_0 : unit root (nonstationary).

2. Trend Modeling and Forecasting - Linear, quadratic, log-linear models. - Holt-Winters smoothing. - Model selection: AIC, BIC. - Forecasting: point, interval, density. - R code samples: `adf.test(series)` `Box.test(series, type="Ljung-Box")`

3. Seasonality - Additive vs multiplicative. - Seasonal dummy models. - STL decomposition. - Tests for remaining seasonality using `tsdisplay`.

4. Cycles & ARMA Models - Wold decomposition → motivation for ARMA. - $AR(p)$: use PACF to identify lag. - $MA(q)$: use ACF to identify lag. - $ARMA(p,q)$ estimation.

5. ARCH & GARCH - Volatility clustering. - $ARCH(q)$: σ^2 depends on past squared errors. - $GARCH(p,q)$: includes lagged variances. - Stationarity: $\alpha + \beta < 1$. - Long-run variance formula. - R code: `garchFit(~ garch(1,1), data=returns)`

6. Forecast Evaluation - Loss functions: MSE, MAE, MAPE. - Mincer-Zarnowitz regression: $y = \alpha + \beta f$; H_0 : $\alpha=0$, $\beta=1$. - Diebold-Mariano: H_0 : equal predictive accuracy.

7. Cointegration & ECM - Engle-Granger procedure. - Residual ADF test. - ECM: $\Delta y = \alpha + \beta \Delta x + \lambda u_{t-1}$ - λ = speed of adjustment. - Structural breaks (Gregory-Hansen).

8. State Space & Kalman Filter - Components: state/transition, measurement. - Q = state noise variance. - R = measurement noise variance. - Kalman gain interpretation. - Use for time-varying betas.

9. Markov Chains & MCMC - Irreducible, aperiodic, ergodic. - Stationary distribution solving $\pi = \pi P$. - Monte Carlo simulation basics. - Metropolis-Hastings. - Gibbs sampler.

Chapter-by-Chapter Summaries (Condensed)

Lecture 1–3: Fundamentals, trend modeling, model selection (AIC/BIC). Lecture 4: White noise, randomness, lag operator. Lecture 5: Seasonality models, forecasting. Lecture 6: Cycles, MA models, invertibility. Lecture 7–9: Full forecasting models, recursive estimation, ARIMA. Lecture 10–12: Forecast evaluation, DM, MZ tests. Lecture 14: ARCH/GARCH models. Lecture 15: Cointegration, ECM. Lecture 16: Kalman filter, state-space. Lecture 17: Monte Carlo, Markov chains, MCMC.